Protocol

Hashable

A type that can be hashed into a `Hasher` to produce an integer hash value.

iOS 8.0+ | iPadOS 8.0+ | Mac Catalyst 13.0+ | macOS 10.10+ | tvOS 9.0+ | visionOS 1.0+ | watchOS 2.0+

```
protocol Hashable : Equatable, ~Copyable, ~Escapable
```

Mentioned in

Adopting Common Protocols

Overview

You can use any type that conforms to the `Hashable` protocol in a set or as a dictionary key. Many types in the standard library conform to `Hashable`: Strings, integers, floating-point and Boolean values, and even sets are hashable by default. Some other types, such as optionals, arrays and ranges automatically become hashable when their type arguments implement the same.

Your own custom types can be hashable as well. When you define an enumeration without associated values, it gains `Hashable` conformance automatically, and you can add `Hashable` conformance to your other custom types by implementing the `hash(into:)` method. For structs whose stored properties are all `Hashable`, and for enum types that have all-`Hashable` associated values, the compiler is able to provide an implementation of `hash(into:)` automatically.

Hashing a value means feeding its essential components into a hash function, represented by the `Hasher` type. Essential